

Auswertung_Bachelorarbeit

```
df <- read.csv("results_run_zs_1.csv", sep = ",")
df
```

	key
1	XD-1005
2	STS-3628
3	RCP-566
4	MOD-261
5	IDE-978
6	DATAREST-407
7	DATA CASS-119
8	MDEV-15187
9	MDEV-8651
10	MXS-1112
11	MDEV-5156
12	CONPY-114
13	UPM-2745
14	UPM-2663
15	UPM-2562
16	UPM-370
17	PL-411
18	AMKT-929
19	AG-624
20	AC-1262

1	UI: User should be able to filter the list of executions on the execution
2	buttons on new JavaFX-based dashboard don't work
3	Hide splash screen before showing main application window
4	Change behaviour of NotificationImpl.hasMessage()
5	No relevant help contents of Spring Project
6	Deleting query methods (starting with removeBy... or deleteBy...) must not support HQL
7	Umlauts (and probably other special characters) are not saved correctly

8 MariaDB crashes when running a stored pr
9 this repository is s
10 Broken link in
11 Long semaphore wai
12 Can't add connection
13 Atlassian Marketplace in Confluence Should Allow Users to List Installed
14 request total is not updated after making a
15 Warn user when updating a plugin from a free plugin version to a marketplace paid plugin v
16 Add title next to Install tab drop
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19 Sometime the mouse pointer changes to hand icon inside the drop down menu of
20 Lots 'Unexpected host info response 503' when add-on is in

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8 I am in the process of testing our dataset for a migration from MariaDB 10.0.33 to MariaD
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	original_priority	predicted_priority	business_value	urgency	impact	risk
1	Minor	Minor	Medium	Low	Low	Low
2	Blocker	Major	Medium	High	Medium	Low
3	Minor	Minor	Low	Low	Low	Low
4	Trivial	Minor	Low	Low	Low	Low
5	Minor	Minor	Low	Low	Low	Low
6	Major	Critical	Medium	High	High	High
7	Blocker	Major	Medium	Low	Medium	High
8	Critical	Critical	High	High	High	High

9	Critical	Critical	Medium	High	High	Low
10	Trivial	Minor	Low	Low	Low	Low
11	Blocker	Critical	Medium	High	High	High
12	Major	Critical	Medium	High	High	High
13	Minor	Minor	Medium	Low	Low	Low
14	Trivial	Major	Medium	Low	Low	Low
15	Major	Minor	Medium	Low	Low	Low
16	Major	Minor	Low	Low	Low	Low
17	Blocker	Critical	Medium	High	Medium	High
18	Critical	Minor	Low	Medium	Low	Low
19	Trivial	Minor	Low	Low	Low	Low
20	Critical	Critical	Medium	High	High	High

	estimated_effort
1	Low
2	Medium
3	Low
4	Low
5	Low
6	Low
7	Medium
8	Medium
9	Medium
10	Low
11	Medium
12	Medium
13	Low
14	Medium
15	Low
16	Low
17	Low
18	Low
19	Low
20	Medium

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The issue has low business value
The issue poses a significant risk

The issue is blocking the upgrade process and

```

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11         The issue causes server crashes which can lead to data loss and system downtime
12                                                     The issue is l
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20 The issue is causing significant user frustration and potentially damaging the user exper
    inference_time
1         4.237
2         4.387
3         4.509
4         3.741
5         4.911
6         4.774
7         4.770
8         5.680
9         5.096
10        4.142
11        5.609
12        4.809
13        4.954
14        4.506
15        4.973
16        4.584
17        4.019
18        4.726
19        4.661
20        5.780

```

```

df_zs <- df[, c("key", "original_priority", "predicted_priority")]
df_zs

```

	key	original_priority	predicted_priority
1	XD-1005	Minor	Minor
2	STS-3628	Blocker	Major
3	RCP-566	Minor	Minor
4	MOD-261	Trivial	Minor
5	IDE-978	Minor	Minor

6	DATAREST-407	Major	Critical
7	DATAACASS-119	Blocker	Major
8	MDEV-15187	Critical	Critical
9	MDEV-8651	Critical	Critical
10	MXS-1112	Trivial	Minor
11	MDEV-5156	Blocker	Critical
12	CONPY-114	Major	Critical
13	UPM-2745	Minor	Minor
14	UPM-2663	Trivial	Major
15	UPM-2562	Major	Minor
16	UPM-370	Major	Minor
17	PL-411	Blocker	Critical
18	AMKT-929	Critical	Minor
19	AG-624	Trivial	Minor
20	AC-1262	Critical	Critical

```

priority_map <- c(
  "Trivial" = 1,
  "Minor" = 2,
  "Major" = 3,
  "Critical" = 4,
  "Blocker" = 5
)

df_zs$original_num <- priority_map[df_zs$original_priority]
df_zs$predicted_num <- priority_map[df_zs$predicted_priority]
df_zs

```

	key	original_priority	predicted_priority	original_num	predicted_num
1	XD-1005	Minor	Minor	2	2
2	STS-3628	Blocker	Major	5	3
3	RCP-566	Minor	Minor	2	2
4	MOD-261	Trivial	Minor	1	2
5	IDE-978	Minor	Minor	2	2
6	DATAREST-407	Major	Critical	3	4
7	DATAACASS-119	Blocker	Major	5	3
8	MDEV-15187	Critical	Critical	4	4
9	MDEV-8651	Critical	Critical	4	4
10	MXS-1112	Trivial	Minor	1	2
11	MDEV-5156	Blocker	Critical	5	4
12	CONPY-114	Major	Critical	3	4
13	UPM-2745	Minor	Minor	2	2

14	UPM-2663	Trivial	Major	1	3
15	UPM-2562	Major	Minor	3	2
16	UPM-370	Major	Minor	3	2
17	PL-411	Blocker	Critical	5	4
18	AMKT-929	Critical	Minor	4	2
19	AG-624	Trivial	Minor	1	2
20	AC-1262	Critical	Critical	4	4

```
cor.test(df_zs$original_num, df_zs$predicted_num, method = "spearman")
```

Warning in cor.test.default(df_zs\$original_num, df_zs\$predicted_num, method = "spearman"): Kann exakten p-Wert bei Bindungen nicht berechnen

Spearman's rank correlation rho

```
data: df_zs$original_num and df_zs$predicted_num
S = 493.5, p-value = 0.002971
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.6289491
```

```
df1 <- read.csv("results_run_lora_1.csv", sep = ",")
df1
```

```
      key
1  XD-1005
2  STS-3628
3  RCP-566
4  MOD-261
5  IDE-978
6  DATAEST-407
7  DATACASS-119
8  MDEV-15187
9  MDEV-8651
10  MXS-1112
11  MDEV-5156
12  CONPY-114
13  UPM-2745
14  UPM-2663
```

15 UPM-2562
16 UPM-370
17 PL-411
18 AMKT-929
19 AG-624
20 AC-1262

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2 buttons on new JavaFX-based dashboard don
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4 Change behaviour of NotificationImpl.hasMess
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6	Major	Critical	Low	High	Medium	High
7	Blocker	Critical	Medium	High	High	High
8	Critical	Critical	Medium	High	High	High
9	Critical	Critical	Medium	High	High	Medium
10	Trivial	Minor	Low	Low	Low	Low
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13	Minor	Minor	Medium	Low	Low	Low
14	Trivial	Critical	Medium	High	High	High
15	Major	Minor	Low	Low	Low	Low
16	Major	Minor	Low	Low	Low	Low
17	Blocker	Critical	Medium	High	High	High
18	Critical	Critical	Low	High	Medium	High
19	Trivial	Minor	Low	Low	Low	Low
20	Critical	Critical	Medium	High	High	High
	estimated_effort					
1	Medium					
2	Medium					
3	Low					
4	Low					
5	Low					
6	Medium					
7	Medium					
8	Medium					
9	Medium					
10	Low					
11	Medium					
12	Medium					
13	Low					
14	Medium					
15	Low					
16	Low					

17	Medium
18	Low
19	Low
20	Medium

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This issue is critical because it directly impacts user experience and system functionali

	inference_time
1	6.828
2	9.236
3	6.622
4	6.354
5	7.628
6	9.540
7	8.129
8	8.377
9	7.486
10	5.987
11	6.031
12	8.246
13	6.956
14	10.523
15	6.057
16	6.987
17	8.803

```

18          7.061
19          6.545
20          8.096

```

```

df_lora <- df1[, c("key", "original_priority", "predicted_priority")]
df_lora

```

	key	original_priority	predicted_priority
1	XD-1005	Minor	Minor
2	STS-3628	Blocker	Critical
3	RCP-566	Minor	Minor
4	MOD-261	Trivial	Minor
5	IDE-978	Minor	Minor
6	DATAREST-407	Major	Critical
7	DATAACASS-119	Blocker	Critical
8	MDEV-15187	Critical	Critical
9	MDEV-8651	Critical	Critical
10	MXS-1112	Trivial	Minor
11	MDEV-5156	Blocker	Critical
12	CONPY-114	Major	Critical
13	UPM-2745	Minor	Minor
14	UPM-2663	Trivial	Critical
15	UPM-2562	Major	Minor
16	UPM-370	Major	Minor
17	PL-411	Blocker	Critical
18	AMKT-929	Critical	Critical
19	AG-624	Trivial	Minor
20	AC-1262	Critical	Critical

```

priority_map <- c(
  "Trivial" = 1,
  "Minor" = 2,
  "Major" = 3,
  "Critical" = 4,
  "Blocker" = 5
)

df_lora$original_num <- priority_map[df_lora$original_priority]
df_lora$predicted_num <- priority_map[df_lora$predicted_priority]
df_lora

```

	key	original_priority	predicted_priority	original_num	predicted_num
--	-----	-------------------	--------------------	--------------	---------------

1	XD-1005	Minor	Minor	2	2
2	STS-3628	Blocker	Critical	5	4
3	RCP-566	Minor	Minor	2	2
4	MOD-261	Trivial	Minor	1	2
5	IDE-978	Minor	Minor	2	2
6	DATA REST-407	Major	Critical	3	4
7	DATA CASS-119	Blocker	Critical	5	4
8	MDEV-15187	Critical	Critical	4	4
9	MDEV-8651	Critical	Critical	4	4
10	MXS-1112	Trivial	Minor	1	2
11	MDEV-5156	Blocker	Critical	5	4
12	CONPY-114	Major	Critical	3	4
13	UPM-2745	Minor	Minor	2	2
14	UPM-2663	Trivial	Critical	1	4
15	UPM-2562	Major	Minor	3	2
16	UPM-370	Major	Minor	3	2
17	PL-411	Blocker	Critical	5	4
18	AMKT-929	Critical	Critical	4	4
19	AG-624	Trivial	Minor	1	2
20	AC-1262	Critical	Critical	4	4

```
cor.test(df_lora$original_num, df_lora$predicted_num, method = "spearman")
```

Warning in cor.test.default(df_lora\$original_num, df_lora\$predicted_num, : Kann exakten p-Wert bei Bindungen nicht berechnen

Spearman's rank correlation rho

```
data: df_lora$original_num and df_lora$predicted_num
S = 384.81, p-value = 0.000445
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
0.7106691
```

```
spearman_results <- data.frame(
  Modell = c("Zero-Shot", "LoRA"),
  Spearman = c(
    cor(df_zs$original_num, df_zs$predicted_num, method = "spearman"),
    cor(df_lora$original_num, df_lora$predicted_num, method = "spearman")
  )
)
```

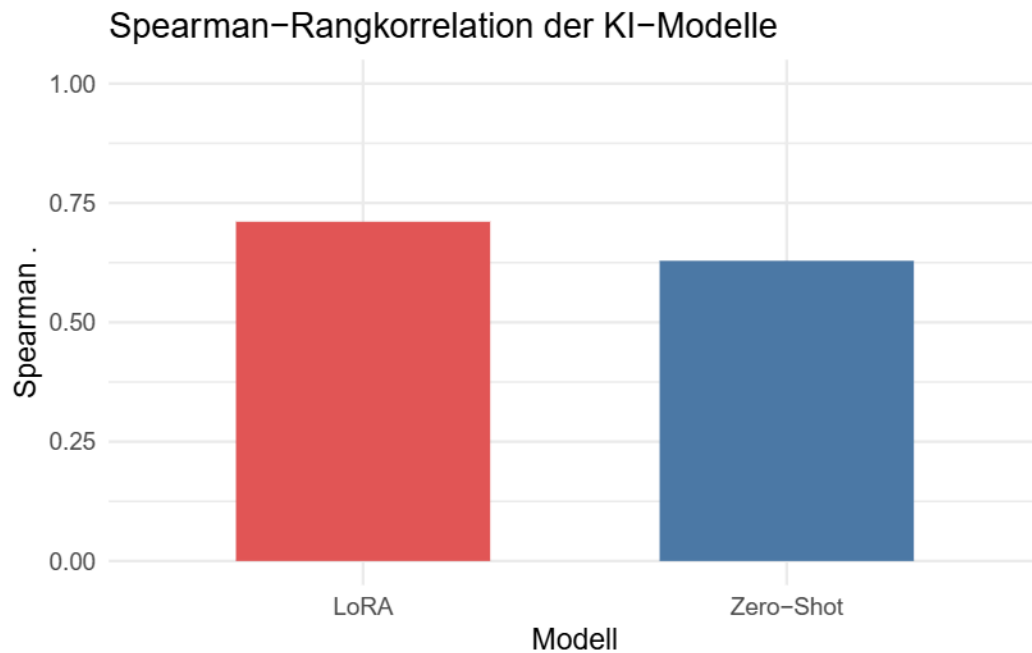
```
)  
)  
  
spearman_results
```

```
      Modell Spearman  
1 Zero-Shot 0.6289491  
2      LoRA 0.7106691
```

```
library(ggplot2)
```

Warning: Paket 'ggplot2' wurde unter R Version 4.4.3 erstellt

```
ggplot(spearman_results, aes(x = Modell, y = Spearman, fill = Modell)) +  
  geom_col(width = 0.6) +  
  ylim(0, 1) +  
  labs(  
    title = "Spearman-Rangkorrelation der KI-Modelle",  
    x = "Modell",  
    y = "Spearman "  
  ) +  
  scale_fill_manual(values = c("Zero-Shot" = "#4E79A7", "LoRA" = "#E15759")) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



```
library(tidyr)
```

Warning: Paket 'tidyr' wurde unter R Version 4.4.3 erstellt

```
library(dplyr)
```

Warning: Paket 'dplyr' wurde unter R Version 4.4.3 erstellt

Attache Paket: 'dplyr'

Die folgenden Objekte sind maskiert von 'package:stats':

filter, lag

Die folgenden Objekte sind maskiert von 'package:base':

intersect, setdiff, setequal, union

```

# Zero-Shot Kriterien
zs_criteria <- df[, c("business_value", "urgency", "impact", "risk", "estimated_effort")] |>
  pivot_longer(everything(), names_to = "Criterion", values_to = "Rating") |>
  mutate(Model = "Zero-Shot")

# LoRA Kriterien
lora_criteria <- df1[, c("business_value", "urgency", "impact", "risk", "estimated_effort")]
  pivot_longer(everything(), names_to = "Criterion", values_to = "Rating") |>
  mutate(Model = "LoRA")

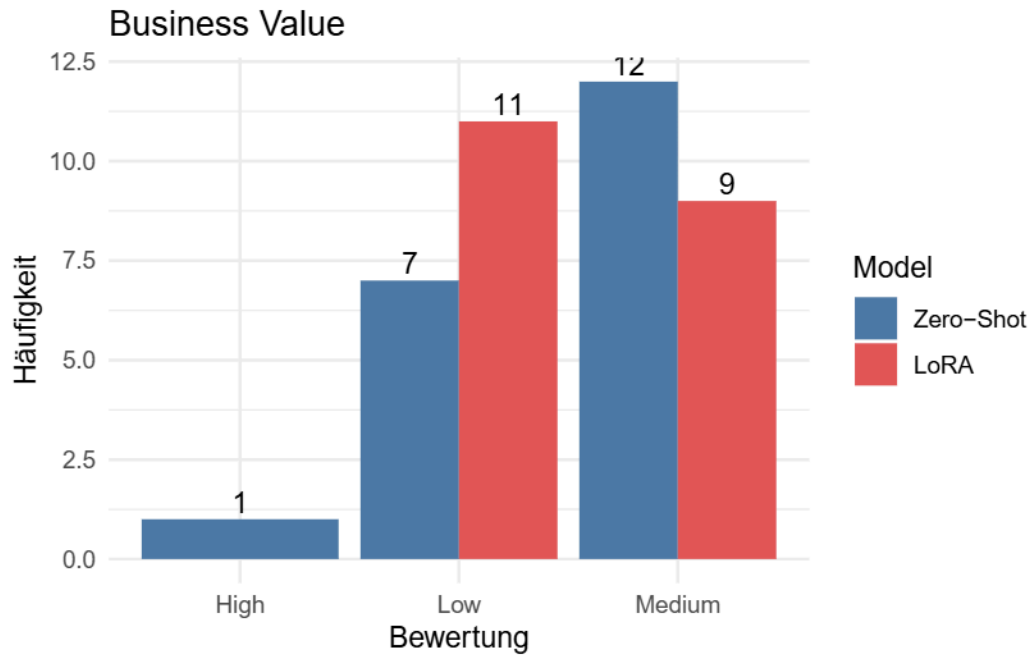
criteria_data <- bind_rows(zs_criteria, lora_criteria)
criteria_data$Model <- factor(criteria_data$Model, levels = c("Zero-Shot", "LoRA"))

library(ggplot2)

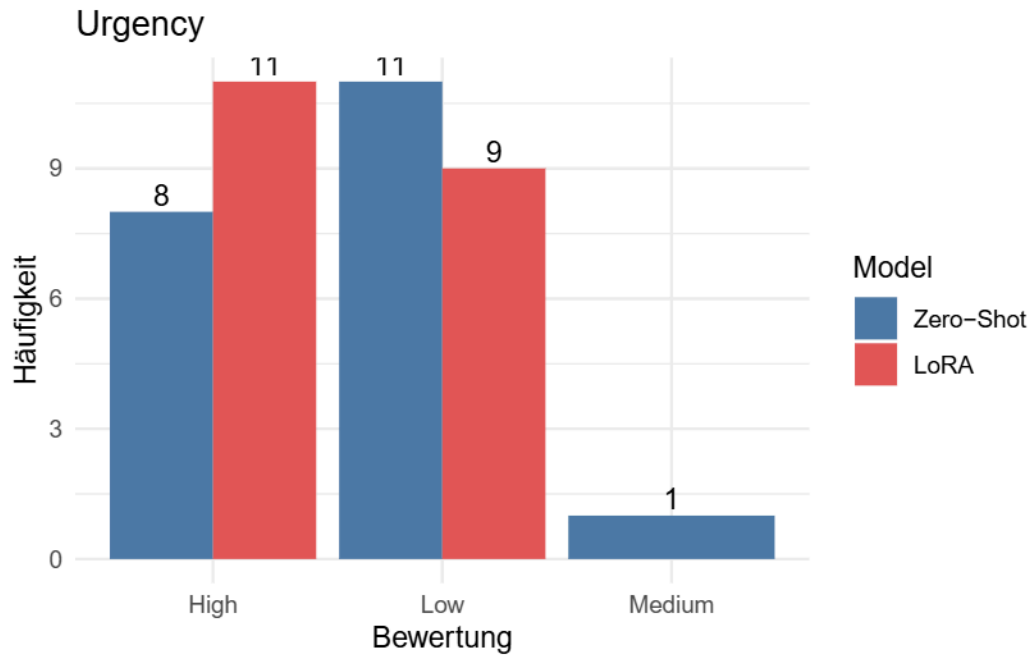
colors <- c("Zero-Shot" = "#4E79A7", "LoRA" = "#E15759")

# Business Value
ggplot(criteria_data[criteria_data$Criterion == "business_value",],
  aes(x = Rating, fill = Model)) +
  geom_bar(position = "dodge") +
  geom_text(stat="count",
    aes(label=after_stat(count)),
    position=position_dodge(width=0.9),
    vjust=-0.3) +
  scale_fill_manual(values = colors) +
  labs(title="Business Value", x="Bewertung", y="Häufigkeit") +
  theme_minimal()

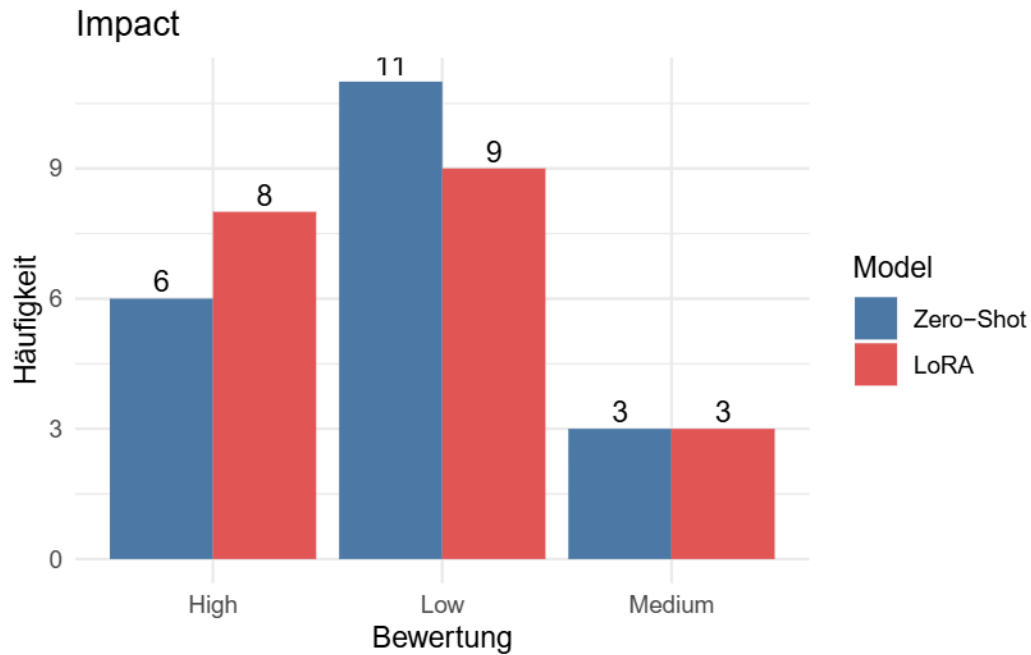
```



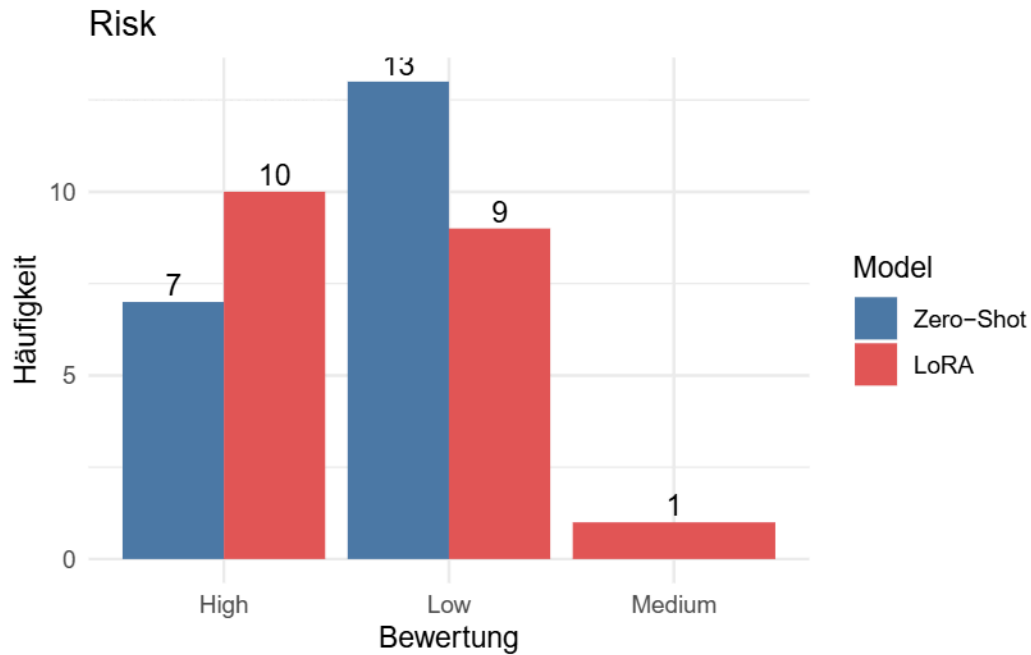
```
# Urgency
ggplot(criteria_data[criteria_data$Criterion == "urgency",],
  aes(x = Rating, fill = Model)) +
  geom_bar(position = "dodge") +
  geom_text(stat="count",
    aes(label=after_stat(count)),
    position=position_dodge(width=0.9),
    vjust=-0.3) +
  scale_fill_manual(values = colors) +
  labs(title="Urgency", x="Bewertung", y="Häufigkeit") +
  theme_minimal()
```



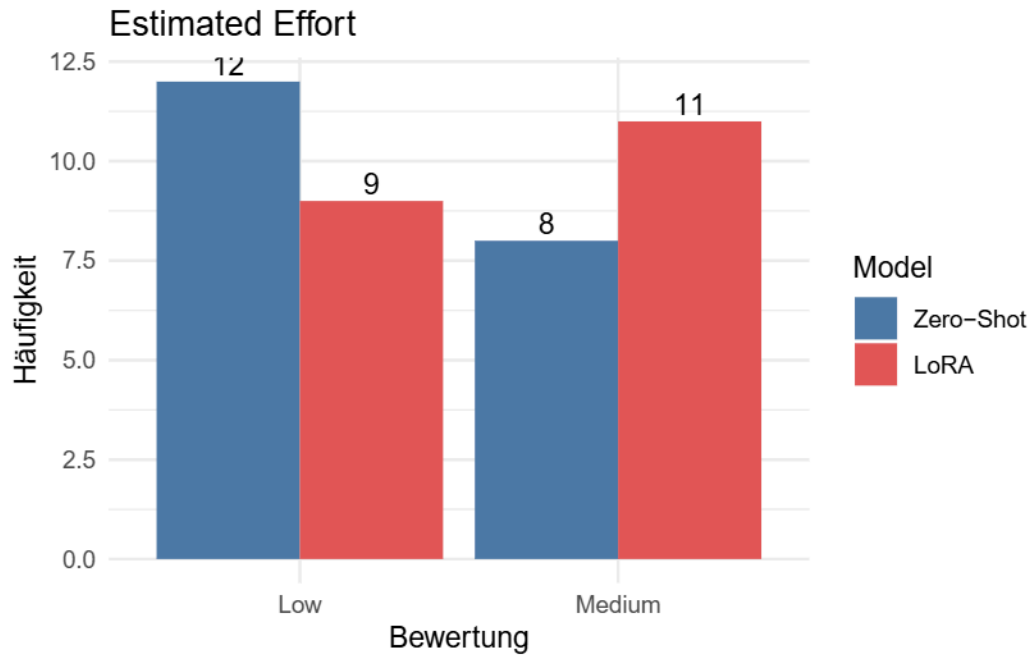
```
# Impact
ggplot(criteria_data[criteria_data$Criterion == "impact",],
  aes(x = Rating, fill = Model)) +
  geom_bar(position = "dodge") +
  geom_text(stat="count",
    aes(label=after_stat(count)),
    position=position_dodge(width=0.9),
    vjust=-0.3) +
  scale_fill_manual(values = colors) +
  labs(title="Impact", x="Bewertung", y="Häufigkeit") +
  theme_minimal()
```



```
# Risk
ggplot(criteria_data[criteria_data$Criterion == "risk",],
  aes(x = Rating, fill = Model)) +
  geom_bar(position = "dodge") +
  geom_text(stat="count",
    aes(label=after_stat(count)),
    position=position_dodge(width=0.9),
    vjust=-0.3) +
  scale_fill_manual(values = colors) +
  labs(title="Risk", x="Bewertung", y="Häufigkeit") +
  theme_minimal()
```



```
# Estimated Effort
ggplot(criteria_data[criteria_data$Criterion == "estimated_effort",],
  aes(x = Rating, fill = Model)) +
  geom_bar(position = "dodge") +
  geom_text(stat="count",
    aes(label=after_stat(count)),
    position=position_dodge(width=0.9),
    vjust=-0.3) +
  scale_fill_manual(values = colors) +
  labs(title="Estimated Effort", x="Bewertung", y="Häufigkeit") +
  theme_minimal()
```



```
# Business Value
cat("\nBusiness Value\n")
```

Business Value

```
bv <- table(criteria_data$Rating[criteria_data$Criterion == "business_value"],
             criteria_data$Model[criteria_data$Criterion == "business_value"])
print(bv)
```

	Zero-Shot	LoRA
High	1	0
Low	7	11
Medium	12	9

```
# Urgency
cat("\nUrgency\n")
```

Urgency

```
urg <- table(criteria_data$Rating[criteria_data$Criterion == "urgency"],
             criteria_data$Model[criteria_data$Criterion == "urgency"])
print(urg)
```

	Zero-Shot	LoRA
High	8	11
Low	11	9
Medium	1	0

```
# Impact
cat("\nImpact\n")
```

Impact

```
imp <- table(criteria_data$Rating[criteria_data$Criterion == "impact"],
             criteria_data$Model[criteria_data$Criterion == "impact"])
print(imp)
```

	Zero-Shot	LoRA
High	6	8
Low	11	9
Medium	3	3

```
# Risk
cat("\nRisk\n")
```

Risk

```
risk <- table(criteria_data$Rating[criteria_data$Criterion == "risk"],
              criteria_data$Model[criteria_data$Criterion == "risk"])
print(risk)
```

	Zero-Shot	LoRA
High	7	10
Low	13	9
Medium	0	1

```
# Estimated Effort
cat("\nEstimated Effort\n")
```

Estimated Effort

```
eff <- table(criteria_data$Rating[criteria_data$Criterion == "estimated_effort"],
             criteria_data$Model[criteria_data$Criterion == "estimated_effort"])
print(eff)
```

	Zero-Shot LoRA	
Low	12	9
Medium	8	11